

FULL STACK COLLEGE PROJECT · PRESENTED TO THE JUDGES

CollabHub

"Find teammates. Build together."

A skill-based teammate matching platform that helps students discover hackathons, projects and competitions, and connects them with the right collaborators using a smart eligibility and match-score engine.

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Why CollabHub?

Students constantly look for hackathon and project teammates on WhatsApp groups and Instagram DMs, with no way to verify if someone actually has the right skills before joining a team. CollabHub solves this by making **skills, skill levels and eligibility** the foundation of every match — so teams form faster and are genuinely capable of doing the work.

The Core Idea

Every user builds a skill profile with self-rated proficiency levels. Every activity (hackathon, project, competition) lists the skills and minimum levels it needs. CollabHub automatically checks eligibility and ranks applicants by a calculated **match score**, turning team formation from guesswork into data.

How a User Moves Through the App



What the Platform Does

MODULE 01**Authentication**

- Register & Login
- Forgot password (optional)
- Secure logout

MODULE 02**User Profile**

- Name, college, department, year, bio, photo
- Skills stored as Skill – Level (e.g. React – Level 5)
- Profile completion percentage

MODULE 03**Activity Feed**

- Card-based feed of Hackathons, Projects, Competitions, Research, Startups, Open Source, Workshops
- Shows required skills, seats left, and creator

MODULE 04**Create Activity**

- Title, description, category, date, mode, max members
- Required skills with minimum skill level
- Application deadline

MODULE 05**Smart Eligibility**

- Auto-checks each required skill against the user's level on Apply
- Instantly shows "Eligible" or the exact gap (e.g. "Need React Level 3, current Level 2")

MODULE 06**Join Requests**

- Creator sees pending applicants with their skill ratings
- One-click Approve / Reject

MODULE 07**Group Chat**

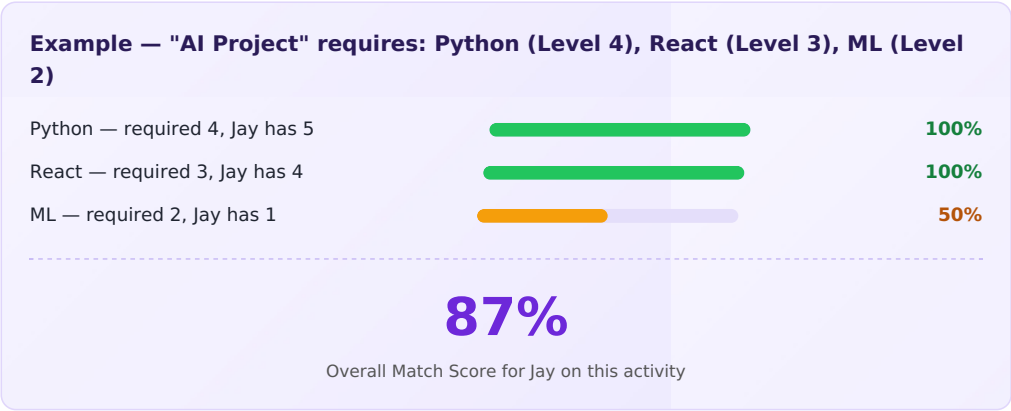
- Unlocked only for approved members
- Messages with sender + timestamp, auto-refresh polling

MODULE 08**Dashboard**

- Activities created & joined
- Pending requests
- Upcoming events at a glance

Smart Match Score Algorithm

Instead of a simple pass/fail eligibility check, CollabHub calculates a **weighted match percentage** between an applicant's skills and an activity's requirements. This lets activity creators sort every applicant by best fit — much closer to how real collaboration platforms and hiring tools work — and gives the project a genuine algorithm to walk through in the viva.



How it's calculated

For each required skill, the applicant's level is compared to the required level (capped at 100% per skill once it meets or exceeds the requirement). The overall score is the average across all required skills, optionally weighted by how critical each skill is to the activity. Activity creators can then sort applicants highest-match first, instead of reviewing requests one by one.

Database Schema

A compact relational schema — seven tables capture users, their skills, activities, requirements, requests and chat.

Users	Skills
id, name, email, password college, department, bio	id, skill_name
User Skills	Activities
user_id, skill_id, level	id, creator_id, title, description category, max_members, deadline
Activity Skills	Join Requests
activity_id, skill_id, required_level	id, activity_id, user_id, status (Pending / Approved / Rejected)
Messages	
id, activity_id, user_id, message, time	

IMPLEMENTATION

Tech Stack

FRONTEND HTML · CSS · JavaScript Simple, dependency-free UI	BACKEND Python — FastAPI / Flask REST APIs	DATABASE MySQL Relational schema
AUTH Sessions / JWT Secure login	CHAT Polling (few sec.) Simpler than WebSockets	HOSTING Any campus / free tier Lightweight stack

BONUS FEATURES

Nice Extras

Search activities by skill	Filter by category
Bookmark activities	Notifications for approvals
Ratings after activity ends	Profile completion %
Badges — Hackathon Winner, Mentor	Match Score sorting for creators

Future Scope

- AI-based teammate recommendations
- Auto-generated resume from completed activities
- Video meeting integration
- Calendar sync
- Real-time chat with WebSockets
- Organization / college pages
- Mobile application

Why This Project Stands Out

- ✓ A real, relatable problem every student has faced — finding capable teammates
- ✓ Demonstrates full-stack skills: auth, relational DB design, CRUD, real-time-feel chat
- ✓ The match-score algorithm gives a concrete, explainable piece of logic for the viva
- ✓ Scoped tightly enough to actually finish, with a clear roadmap to grow further